

MATH1141: Higher Mathematics 1A - Algebra

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April 18, 2026

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Chapter 1

Introduction to Vectors

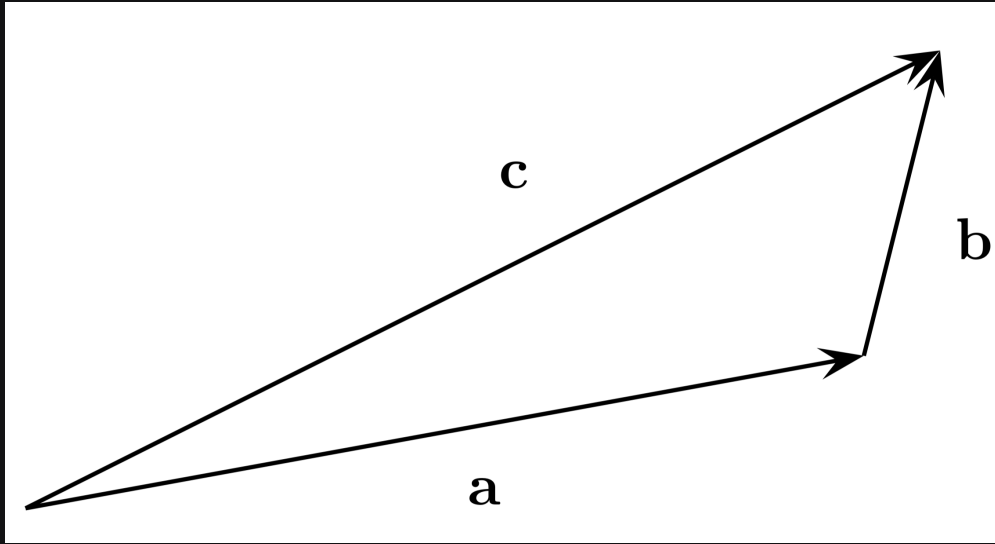
1.1 Vector quantities

- A **scalar** quantity is anything that can be specified by a single number
- A **vector** quantity is specified by both a magnitude and direction

1.1.1 Geometric vectors

- A geometric vector is often represented by an arrow or a directed line segment
- The length of the arrow represents the magnitude of the vector and the arrow points in the direction of the vector
- In these notes, vectors will be denoted as $\mathbf{v}$
- Two vectors are equal if they share the same direction and magnitude. Hence, a translation of a vector would be the same vector.

Vectors $\mathbf{u}$ and $\mathbf{v}$ can be added "head to tail" to produce a new vector $\mathbf{u} + \mathbf{v}$



The zero vector $\mathbf{0}$ has no length and is the only vector with no specific direction. Adding the zero vector to a vector does not change it.

Scalar multiplication is the process of multiplying a vector $\mathbf{v}$ by a scale $\lambda \in \mathbb{R}$. If $\lambda \geq 0$, then $\lambda \mathbf{v}$ denotes the vector with the same direction, but magnitude is scaled by λ , ie. $|\lambda \mathbf{v}| = \lambda |\mathbf{v}|$.

Properties of vector arithmetic

Vector manipulation is defined similarly to regular numbers (the names of these laws should be memorised):

- **Commutative law:** $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$
- **Additive Associative law:** $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
- **Multiplicative Associative law:** $\lambda(\mu \mathbf{v}) = (\lambda\mu)\mathbf{v} := \lambda\mu \mathbf{v}$
- **Distributive law:**
 - Vector $\rightarrow \lambda(\mathbf{v} + \mathbf{w}) = \lambda \mathbf{v} + \lambda \mathbf{w}$
 - Scalar $\rightarrow (\lambda + \mu)\mathbf{v} = \lambda \mathbf{v} + \mu \mathbf{v}$

Two vectors $\mathbf{a}$, $\mathbf{b}$ are parallel if there exists a scalar λ such that either $\mathbf{a} = \lambda \mathbf{b}$ or $\mathbf{b} = \lambda \mathbf{a}$.

Coordinates

To place coordinates on a plane the following must be specified:

- an origin point O
- a pair of vectors $\mathbf{i}$ and $\mathbf{j}$ that both have a unit length and are at right angles to each other (it is conventional to choose $\mathbf{j}$ at $\frac{\pi}{2}$) anticlockwise from $\mathbf{i}$

Every geometric vector $\mathbf{a}$ in the plane can be written as $\mathbf{a} = a_1 \mathbf{i} + a_2 \mathbf{j} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$.

1.2 Vector quantities and $\mathbb{R}^n$

Coordinate vectors can be generalised to be any number of components. Let n be a positive integer. An n -tuple or n -vector is an ordered list of n numbers $a_1, a_2, \dots, a_n$ and can be either written as a row or column vector

The set of all n -tuples is denoted $\mathbb{R}^n$. Thus

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} \in \mathbb{R}^2, \quad \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \in \mathbb{R}^3, \quad \begin{pmatrix} 0 \\ 1 \\ -1 \\ \pi \end{pmatrix} \in \mathbb{R}^4$$

1.2.1 Arithmetic on $\mathbb{R}^n$

Vector addition and scalar multiplication in $\mathbb{R}^n$ is defined coordinatewise, as previously demonstrated in $\mathbb{R}^2$ and $\mathbb{R}^3$.

$$\begin{pmatrix} a_1 \\ a_2 \\ \dots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{pmatrix} := \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \dots \\ a_n + b_n \end{pmatrix}$$

The 'zero element' is $\mathbf{0} = \begin{pmatrix} 0 \\ \dots \\ 0 \end{pmatrix}$, and the negative is given by $-\begin{pmatrix} a_1 \\ \dots \\ a_n \end{pmatrix} = \begin{pmatrix} -a_1 \\ \dots \\ -a_n \end{pmatrix}$

Note that each $\mathbb{R}^n$ is a separate system, eg. a 3-tuple cannot be added to a 7-tuple

The length of a vector $\mathbf{a}$ is defined as $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$

1.2.2 Lines in $\mathbb{R}^n$

A line L in $\mathbb{R}^n$ is determined by:

- a point A on the line, eg. with $\mathbf{a} = \overrightarrow{OA}$
- a direction, given by a non-zero vector $\mathbf{v}$

$$\mathbf{x} = \mathbf{a} + \lambda \mathbf{v}, \quad \lambda \in \mathbb{R}$$

This form is referred to as parametric vector form of the line L .

A line in $\mathbb{R}^n$ is any set of the form

$$\{\mathbf{x} \in \mathbb{R}^n | \mathbf{x} = \mathbf{a} + \lambda \mathbf{v}, \quad \lambda \in \mathbb{R}, \quad \mathbf{v} \neq \mathbf{0}, \quad \mathbf{a} \in \mathbb{R}^n\}$$

1.2.3 Parametric to Cartesian form

Example

Write the line $\mathbf{x} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$ in Cartesian form

$$\text{Let } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 + 2\lambda \\ -1 + \lambda \end{pmatrix}$$

$$\lambda = \frac{x - 1}{2} = y + 1$$

Example

Consider $\mathbf{x} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 + \lambda \end{pmatrix}$$

In this case, $x = 1$.

1.2.4 Cartesian form for lines and planes in $\mathbb{R}^3$

A plane in xyz-space can be described by the equation

$$ax + by + cz = d$$

where not all a, b, c are 0. This is called the Cartesian form for the plane. The Cartesian form for a line in $\mathbb{R}^n$ can be defined as the intersection of two planes ($P_1 \cap P_2$). Usually, this can be written in the form

$$\frac{x - a_1}{v_1} = \frac{y - a_2}{v_2} = \frac{z - a_3}{v_3}$$

for some constants a_i, v_i .

Example

Find the parametric form for $\frac{x-2}{3} = \frac{y+1}{6} = \frac{z-3}{-2}$

$$\lambda = \frac{x-2}{3} = \frac{y+1}{6} = \frac{z-3}{-2}$$

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3\lambda + 2 \\ 6\lambda_1 \\ -2\lambda + 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 6 \\ -2 \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

1.2.5 Parametric to Cartesian form for lines in $\mathbb{R}^3$

Example

Find a Cartesian form for $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \lambda \in \mathbb{R}$

$$\begin{pmatrix} 4\lambda + 1 \\ 5\lambda + 2 \\ 6\lambda + 3 \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\lambda = \frac{x-1}{4} = \frac{y-2}{5} = \frac{z-3}{6}$$

Example

Consider $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 5 \\ 0 \end{pmatrix}$

$$\lambda = \frac{x-1}{4} = \frac{y-2}{5}, \quad z = 3$$

1.2.6 Linear combination

Suppose that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$. A linear combination of these vectors is a vector of the form

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k, \quad \lambda_1, \dots, \lambda_k \in \mathbb{R}$$

1.2.7 Span

The span of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ is the set of all linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$, ie.

$$\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \{\lambda_1 \mathbf{v}_1, \lambda_2 \mathbf{v}_2, \dots, \lambda_k \mathbf{v}_k | \lambda_1, \dots, \lambda_k \in \mathbb{R}\}$$

Note that $\text{span}(\emptyset) = \{\mathbf{0}\}$

1.2.8 Planes in $\mathbb{R}^n$

A plane in $\mathbb{R}^n$ is defined to be a set of the form

$$S = \{\mathbf{a} + \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 | \lambda_1, \lambda_2 \in \mathbb{R}\}$$

where $\mathbf{a}$, $\mathbf{v}_1$ and $\mathbf{v}_2$ are fixed vectors in $\mathbb{R}^n$, and $\mathbf{v}_1$ and $\mathbf{v}_2$ are not parallel.

The expression $\mathbf{x} = \mathbf{a} + \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2$, $\lambda_1, \lambda_2 \in \mathbb{R}$ is a parametric vector form for the plane through $\mathbf{a}$ parallel to the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$.

Example

Find a parametric vector equation for the plane containing the point $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, and line $\mathbf{x} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix}$, $\lambda \in \mathbb{R}$.

$$\text{Let direction } \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$\mathbf{x} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R}$$

Example

Find the Cartesian equation of the plane in $\mathbb{R}^3$: $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$, $\lambda_1, \lambda_2 \in \mathbb{R}$.

Equating coordinates and adding:

$$x + y = 3 + 5\lambda_2$$

$$y + z = 5 + 3\lambda_2$$

$$\text{Hence } 3(x + y) - 5(y + z) = 3(3) - 5(5) = -16$$

$$\therefore -16 = 3x - 2y - 5z$$

Chapter 2

Vector Geometry

2.1 Angles via cosine rule

Let $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB} \in \mathbb{R}^n$ be non-zero. The cosine rule is as follows

$$|\mathbf{b} - \mathbf{a}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}||\mathbf{b}| \cos \theta$$

The formula for the length of a vector gives

$$\sum_i (b_i - a_i)^2 = \sum_i a_i^2 + \sum_i b_i^2 - 2|\mathbf{a}||\mathbf{b}| \cos \theta$$

Cancelling gives $-2|\mathbf{a}||\mathbf{b}| \cos \theta = \sum_i -2a_i b_i$ so

$$\cos \theta = \frac{\sum_i a_i b_i}{|\mathbf{a}||\mathbf{b}|}$$

θ is solvable as long as RHS lies in $[-1, 1]$ (Cauchy-Schwarz theorem)

The dot product of two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ is defined to be

$$\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$$

Hence, the angle between two non-zero vectors $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB} \in \mathbb{R}^n$ is

$$\theta = \cos^{-1} \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} \right)$$

Properties of the dot product

Suppose $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$. Then

- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$

- $\mathbf{a} \cdot (\lambda \mathbf{b}) = (\lambda \mathbf{a}) \cdot \mathbf{b} = \lambda(\mathbf{a} \cdot \mathbf{b})$
- $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$
- $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2 \geq 0$

2.1.1 Cauchy-Schwarz theorem

The Cauchy-Schwarz theorem is as follows

$$-|\mathbf{a}||\mathbf{b}| \leq \mathbf{a} \cdot \mathbf{b} \leq |\mathbf{a}||\mathbf{b}|$$

Proof

The inequality holds when $\mathbf{b} = \mathbf{0}$ so it is assumed $\mathbf{b} \neq \mathbf{0}$. Consider the real function of (the real variable) λ

$$q(\lambda) = |\mathbf{a} - \lambda \mathbf{b}|^2 \geq 0$$

$$q(\lambda) = (\mathbf{a} - \lambda \mathbf{b}) \cdot (\mathbf{a} - \lambda \mathbf{b})$$

$$= |\mathbf{a}|^2 - 2\lambda \mathbf{a} \cdot \mathbf{b} + \lambda^2 |\mathbf{b}|^2$$

The discriminant of this quadratic function of λ must be non-positive, hence

$$0 \geq (-2\mathbf{a} \cdot \mathbf{b})^2 - 4|\mathbf{a}|^2|\mathbf{b}|^2 = 4(\mathbf{a} \cdot \mathbf{b})^2 - 4|\mathbf{a}|^2|\mathbf{b}|^2 \implies (\mathbf{a} \cdot \mathbf{b})^2 \leq |\mathbf{a}|^2|\mathbf{b}|^2$$

2.2 Orthogonality

Two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ are said to be orthogonal if $\mathbf{a} \cdot \mathbf{b} = 0$, ie. the angle between them is $\frac{\pi}{2}$

Example

Consider $\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 1 \\ 1 \\ 2 \end{pmatrix}$

$$\mathbf{a} \cdot \mathbf{b} = (1 \times -2) + (1 \times 1) + (-1 \times 1) + (1 \times 2) = 0$$

therefore, $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.

If $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ are orthogonal then

$$|\mathbf{a} + \mathbf{b}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2$$

2.2.1 Orthonormal sets of vectors

The vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ form an orthogonal set if they are mutually orthogonal. Furthermore, if they all have length 1, the vectors are orthonormal.

Equivalently, $\mathbf{v}_1, \dots, \mathbf{v}_k$ are orthonormal if

$$v_i \cdot v_j = \delta_{ij} := \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

[δ_{ij} is called the Kronecker delta symbol]

Arithmetic properties of the cross product

For $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$ and $\lambda \in \mathbb{R}$

- $\mathbf{a} \times \mathbf{a} = \mathbf{0}$
- $\times$ is distributive:
 - $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$
 - $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \times \mathbf{c}) + (\mathbf{b} \times \mathbf{c})$
- $(\lambda \mathbf{a}) \times \mathbf{b} = \lambda(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (\lambda \mathbf{b})$
- Since $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$, $\times$ is not commutative
- The cross product is not associative

The magnitude of $\mathbf{a} \times \mathbf{b}$

Suppose that $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ are nonzero, with angle θ between them. Then $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin \theta$. Since $\theta \in [0, \pi]$ in $\mathbb{R}^3$, therefore $\sin \theta \geq 0$. Thus, it is sufficient that

$$|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2 |\mathbf{b}|^2 \sin^2 \theta$$

Areas of parallelograms via cross product

The area of a parallelogram is

$$A = \text{base} \times \text{perp height} = |\mathbf{a}||\mathbf{b}| \sin \theta$$

Direction of $\mathbf{a} \times \mathbf{b}$

Row swapping of determinants gives the following

- $\mathbf{a}(\mathbf{b} \times \mathbf{c}) = -\mathbf{a}(\mathbf{c} \times \mathbf{b}) = \mathbf{c}(\mathbf{a} \times \mathbf{b})$
- $\mathbf{a}(\mathbf{a} \times \mathbf{b}) = 0 = \mathbf{a}(\mathbf{b} \times \mathbf{b})$

The vector $\mathbf{a} \times \mathbf{b}$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$

Parametric to Cartesian form via point-normal

Example

Find a point-normal, and hence a Cartesian form for the plane $\mathbf{x} = \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} +$

$$\lambda_2 \begin{pmatrix} 4 \\ 0 \\ 3 \end{pmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R}$$

A point on the line is $\mathbf{a} = \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}$

A vector normal to the plane is $\mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \times \begin{pmatrix} 4 \\ 0 \\ 3 \end{pmatrix}$

Chapter 3

Complex Numbers

3.0.1 Fields

A field is a non empty set for which a rule for addition and multiplication are defined. The field must define values for the following axioms

1. **Closure under Addition.** If $x, y \in \mathbb{F}$ then $x + y \in \mathbb{F}$.
2. **Associative Law of Addition.** $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}$.
3. **Commutative Law of Addition.** $x + y = y + x$ for all $x, y \in \mathbb{F}$.
4. **Existence of a Zero.** There exists an element of $\mathbb{F}$ (usually written as 0) such that $0 + x = x + 0 = x$ for all $x \in \mathbb{F}$.
5. **Existence of a Negative.** For each $x \in \mathbb{F}$, there exists an element $w \in \mathbb{F}$ (usually written as $-x$) such that $x + w = w + x = 0$.
6. **Closure under Multiplication.** If $x, y \in \mathbb{F}$ then $xy \in \mathbb{F}$.
7. **Associative Law of Multiplication.** $x(yz) = (xy)z$ for all $x, y, z \in \mathbb{F}$.
8. **Commutative Law of Multiplication.** $xy = yx$ for all $x, y \in \mathbb{F}$.
9. **Existence of a One.** There exists a non-zero element of $\mathbb{F}$ (usually written as 1) such that $x1 = 1x = x$ for all $x \in \mathbb{F}$.
10. **Existence of an Inverse for Multiplication.** For each non-zero $x \in \mathbb{F}$, there exists an element w of $\mathbb{F}$ (usually written as $1/x$ or x^{-1}) such that $xw = wx = 1$.
11. **Distributive Law.** $x(y + z) = xy + xz$ for all $x, y, z \in \mathbb{F}$.
12. **Distributive Law.** $(x + y)z = xz + yz$ for all $x, y, z \in \mathbb{F}$.

3.0.2 Complex numbers

A complex number is a formal expression of the form $a + bi$ for some $a, b \in \mathbb{R}$. In particular, two such numbers $a + bi, a' + b'i$ are equal iff $a = a', b = b'$ as real numbers. The real part of $a + bi$ is $\text{Re}(a + bi) =$

a and the imaginary part is $\text{Im}(a + bi) = b$.

Arithmetic of complex numbers

Given the complex numbers $a + bi, a' + b'i$, addition and multiplication is defined as follows

$$(a + bi) + (a' + b'i) = (a + a') + (b + b')i$$

$$(a + bi)(a' + b'i) = (aa' - bb') + (ab' + a'b)i$$

The complex set $\mathbb{C}$ extends the real number system since complex numbers of form $a + 0i$ add and multiply the same way as real numbers. It contains the number i satisfying $i^2 = -1$

Cartesian form

A complex number z written in the form $a + bi$ with $a, b \in \mathbb{R}$ is called the Cartesian form

Example

Express $\frac{1+i}{1-i} - \frac{1-i}{1+i}$ in Cartesian form

$$\begin{aligned} \frac{1+i}{1-i} - \frac{1-i}{1+i} &= \frac{(1+i)^2 - (1-i)^2}{(1-i)(1+i)} \\ &= \frac{(1+2i+i^2) - (1-2i+i^2)}{1^2 + 1^2} \\ &= \frac{4i}{2} \\ &= 2i \end{aligned}$$

Conjugation

The conjugate of a complex number $z = a + bi$ is defined as $\bar{z} = a - bi$

- z is real iff $\bar{z} = z$
- $\overline{\bar{z}} = z$
- $\overline{z + w} = \bar{z} + \bar{w}$ and $\overline{z - w} = \bar{z} - \bar{w}$
- $\overline{zw} = \bar{z} \bar{w}$ and $\overline{\frac{z}{w}} = \frac{\bar{z}}{\bar{w}}$
- $\text{Re}(z) = \frac{1}{2} (z + \bar{z})$ and $\text{Im}(z) = \frac{1}{2i} (z - \bar{z})$

Example

Show that for any $z \in \mathbb{C}$, $(i + 5)z - (i - 5)\bar{z}$ is real.

$$\begin{aligned}\text{Let } w &= (i + 5)z - (i - 5)\bar{z} \\ \bar{w} &= \overline{(i + 5)z - (i - 5)\bar{z}} \\ &= (5 - i)\bar{z} - (-i - 5)z \\ &= (i - 5)\bar{z} + (i + 5)z \\ &= w\end{aligned}$$

3.1 The Argand diagram

Similarly to how real numbers can be represented on a plane, complex numbers can be represented on the complex plane (the Argand diagram).

$z = a + bi$ is represented by the point with coords $(a, b) = (\text{Re}(z), \text{Im}(z))$

Adding complex numbers by adding real and imaginary parts (ie. coordinatewise) can be represented geometrically by the addition of vectors.

3.1.1 Polar form

Given the Cartesian form of a complex number $z = x + iy$, $x, y \in \mathbb{R}$, polar coordinates for $z = (r, \theta)$ can be found

$$z = r \cos \theta + (r \sin \theta) i$$

Let $z = x + iy$, $x, y \in \mathbb{R}$

- The **modulus** of z is defined as $|z| = r = \sqrt{x^2 + y^2}$ so $z\bar{z} = |z|^2$
- If $z \neq 0$, an **argument** for z is any $\theta = \arg z$ so that $\tan \theta = \frac{y}{x}$
- Furthermore, $\theta =: \text{Arg } z$ is the **principal argument** if $\theta \in (-\pi, \pi]$

Example

Find the modulus and principal argument of $-1 - \sqrt{3}i$.

$$|-1 - \sqrt{3}i| = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\arctan \frac{-\sqrt{3}}{-1} = \arctan \sqrt{3} = \frac{\pi}{3}$$

$$\text{Arg}(-1 - \sqrt{3}i) = \frac{\pi}{3} - \pi = -\frac{2\pi}{3}$$

3.1.2 Euler's formula

For $\theta \in \mathbb{R}$, it is defined $e^{i\theta} = \cos \theta + i \sin \theta$. This formula is reasonable by

- $e^{i\theta_1} e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$
- (De Moivre's theorem) For $n \in \mathbb{Z}$, $(e^{i\theta})^n = e^{in\theta}$
- $\frac{d}{d\theta}(e^{i\theta}) = ie^{i\theta}$

Arithmetic of polar forms

The polar form of z is $z = re^{i\theta}$ where $r = |z|$ and θ is an argument of z . The above formulae give

$$r_1 e^{i\theta_1} r_2 e^{i\theta_2} = (r_1 r_2) e^{i(\theta_1 + \theta_2)} \quad , \quad (re^{i\theta})^{-1} = r^{-1} e^{-i\theta}$$

ie. multiplication of complex numbers can be achieved by multiplying the moduli and adding the arguments. Inverting a complex number inverts the modulus and negates the argument.

Example

Find the exact value of $\text{Arg} \frac{1+i}{1+\sqrt{3}i}$

$$\begin{aligned} \arg \frac{1+i}{1+\sqrt{3}i} &= \arg(1+i) - \arg(1+\sqrt{3}i) \\ &= \frac{\pi}{4} - \frac{\pi}{3} \end{aligned}$$

$$\text{Arg} \frac{1+i}{1+\sqrt{3}i} = -\frac{\pi}{12}$$

Example

Let $z \in \mathbb{C}$ have $|z| = 1$. Show that $w = \frac{i-z}{i+z}$ is purely imaginary in the sense that $\operatorname{Re}(w) = 0$. Interpret the result geometrically.

$$\operatorname{Re}(w) = \frac{1}{2}(w + \bar{w})$$

$$\begin{aligned}\bar{w} &= \overline{\frac{i-z}{i+z}} \\ &= \frac{-i - \bar{z}}{-i + \bar{z}} \times \frac{iz}{iz} \\ &= \frac{z - i|z|^2}{z + i} \\ &= -w\end{aligned}$$

Therefore, w is purely imaginary

Example

Find the complex square roots $\pm z$ of $16 - 30i$

Let $z = a + bi$, $a, b \in \mathbb{R}$ such that $z^2 = 16 - 30i$

$$\begin{aligned}\therefore 16 - 30i &= (a + bi)^2 = a^2 + 2abi - b^2 \\ &= a^2 - b^2 + 2abi\end{aligned}$$

$$\text{Real part gives: } 16 = a^2 - b^2$$

$$\text{Imaginary part gives: } -30 = 2ab \Leftrightarrow -15 = ab$$

$$z = \pm(5 - 3i)$$

Example

Solve $z^2 + (1 - i)z + (-4 + 8i) = 0$

$$z = \frac{-(1 - i) \pm \sqrt{(1 - i)^2 - 4(1)(-4 + 8i)}}{2(1)}$$

$$\Delta = b^2 - 4ac = (1 - i)^2 - 4(1)(-4 + 8i) = 1^2 + 2i + i^2 + 16 - 32i = 16 - 30i$$

From above, $\sqrt{\Delta} = \pm(5 - 3i)$

$$z = \frac{-(1 + i) \pm (5 - 3i)}{2} = \frac{4 - 4i}{2} \text{ or } \frac{-6 + 2i}{2}$$

$$= 2 - 2i, \quad -3 + i$$

3.1.3 Powers of complex numbers

Example

Find $w = (1 + i)^{18}$

$$\text{Let } z = 1 + i = \sqrt{2}e^{i\frac{\pi}{4}}$$

$$\begin{aligned} w &= z^{18} \\ &= (\sqrt{2}e^{i\frac{\pi}{4}})^{18} \\ &= 2^9 e^{i\frac{9\pi}{2}} \\ &= 2^9 e^{i\frac{\pi}{2}} \\ &= 2^9 i \end{aligned}$$

3.1.4 Roots of complex numbers

Example

Solve $z^4 = i$.

Solve in polar form for $z = re^{i\theta}$, $\pi < \theta \leq \pi$

Equate moduli: $1 = |i| = |z^4| = |z|^4$, $\therefore r = 1$

Equate arguments: $\arg(z^4) = \arg i$

$$4 \operatorname{Arg}(z) = \frac{\pi}{2}$$

... (got lazy, answer trivial)

3.1.5 Expressing trigonometric polynomials as polynomials in $\cos \theta$, $\sin \theta$

A trigonometric polynomial is a linear combination of functions of the form $\cos n\theta$, $\sin n\theta$.

Example

Use De Moivre's theorem to show $\cos(3\theta) = 4 \cos^3 \theta - 3 \cos \theta$

$$\begin{aligned}\cos 3\theta &= \operatorname{Re}(e^{i3\theta}) \\&= \operatorname{Re}(\cos \theta + i \sin \theta)^3 \\&= \operatorname{Re}(\cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta) \\&= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\&= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) \\&= 4 \cos^3 \theta - 3 \cos \theta\end{aligned}$$

3.2 Triangle inequality

Example

$$\begin{aligned}|z + w|^2 &= (z + w)\overline{(z + w)} \\&= (z + w)(\bar{z} + \bar{w}) \\&= z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w} \\&= |z|^2 + z\bar{w} + \overline{z\bar{w}} + |w|^2 \\&= |z|^2 + 2 \operatorname{Re}(z\bar{w}) + |w|^2 \\&\leq |z|^2 + 2|z\bar{w}| + |w|^2 \\&= (|z| + |w|)^2\end{aligned}$$

3.3 Complex polynomials

A function $p : \mathbb{C} \rightarrow \mathbb{C}$ of the form

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

is called a complex polynomial.

A root of a polynomial p is any $\alpha \in \mathbb{C}$ such that $p(\alpha) = 0$.

Gauss' theorem states that every complex polynomial of degree at least one has a root $\alpha \in \mathbb{C}$.

3.3.1 Factorising polynomials

Let p, q be complex polynomials of degree at least 1. Then q is a factor of p if there is a polynomial r such that $p = qr$.

3.4 Roots of real polynomials

A polynomial is real if the coefficients are real.

Suppose that α is a root of a real polynomial p . Then $\bar{\alpha}$ is also a root of p . If $(z - \alpha)$ and $(z - \bar{\alpha})$ are factors of p and $\alpha \notin \mathbb{R}$, then p has a quadratic factor

Chapter 4

Linear equations

4.1 Linear equations

A linear equation in n unknowns $x_1, x_2, \dots, x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

4.2 Point normal form for lines in $\mathbb{R}^2$

Just as there are point normal forms for planes in $\mathbb{R}^3$, there are point normal forms for lines in $\mathbb{R}^2$. The solution set of two simultaneous equations is the intersection of the lines that represent each equation.

4.3 Intersecting 3 planes in $\mathbb{R}^3$

The produce of systematically solving simultaneous equations and understanding the set is called Gaussian elimination.

To do so, the system of m linear equations in n variables

$$\begin{cases} a_1x_1 + a_2x_2 + \dots + a_nx_n = b_1 \\ a_2x_1 + a_2x_2 + \dots + a_nx_n = b_2 \\ \dots \\ a_mx_1 + a_m2x_2 + \dots + a_mnx_n = b_m \end{cases}$$

is abbreviated to **augmented matrix form**

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right)$$

Example

Write the augmented matrix form for the system

$$\begin{cases} 6x_1 - 2x_2 + 5x_3 = 7 \\ 3x_1 + x_2 + 2x_3 = 0 \end{cases}$$

Solution

$$\left(\begin{array}{ccc|c} 6 & -2 & 5 & 7 \\ 3 & 1 & 2 & 0 \end{array} \right)$$

4.4 Matrices

A matrix is a rectangular array of numbers, such as

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 7 & 5 \end{pmatrix}.$$

This example would be a 2×3 matrix since it has 2 rows and 3 columns.

The augmented matrix consists of the coefficient matrix (on the left of the line) augmented by the right side vector.

Any system of simultaneous equations can be written as a single vector equation.

Example

Write a vector equation equivalent to the following augmented matrix.

$$\left(\begin{array}{cc|c} 1 & 0 & -2 \\ 2 & e^2 & \pi \end{array} \right)$$

$$x_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ e^2 \end{pmatrix} = \begin{pmatrix} -2 \\ \pi \end{pmatrix}$$

A system of linear equations is consistent if it has at least one solution and inconsistent otherwise.

The systems of linear equations $(A|\mathbf{b})$, $(A'|\mathbf{b}')$ are equivalent if they have exactly the same set of solutions.

4.4.1 Back-substitution

We can solve systems of linear equations if they are in row echelon form as in the following

$$\left(\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 8 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right).$$

The solution can be written in the vector form

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 3 - 2\lambda \\ \lambda \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix},$$

hence representing the solution set as a line in $\mathbb{R}^4$.

4.4.2 Gaussian elimination

EROs

Addition

1. Add/subtract rows to get zeroes in the first column of every row, except for the first row
2. Continue this process for each row to form row echelon form

Pivot (just swap the rows)

4.5 Solving simultaneous equations: functional viewpoint

We often solve equations like $\sin x = \frac{1}{2}$, or more generally $f(x) = c$ for some $f : \mathbb{R} \rightarrow \mathbb{R}$ and constant $c \in \mathbb{R}$. This can be used to solve simultaneous equations if we allow vector valued functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$

Consider $a_1, \dots, a_n \in \mathbb{R}$ and the function $I : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$I(x) = a_1x_1 + \dots + a_nx_n.$$

This is a scalar valued linear function with vector inputs. $I(x) = b$ is a linear equation.

Matrix-vector product

Calculate

$$\begin{pmatrix} 1 & 3 \\ 1 & 2 \\ 4 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} \\ = \begin{pmatrix} 1 \times 2 + 3(-1) \\ 1 \times 2 + 2(-1) \\ 4 \times 2 + 0(-1) \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 8 \end{pmatrix}$$

- Distributive law applies

4.6 Linearity

A function

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

is linear if for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \lambda \in \mathbb{R}$ we have

$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}), \quad T(\lambda \mathbf{x}) = \lambda T(\mathbf{x})$$

Example

$T \begin{pmatrix} x \\ y \end{pmatrix} = x^2 + y$ is not linear since

if $\lambda = 2$ and $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$ then

$$T \left(2 \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right) = T \left(\begin{pmatrix} 6 \\ 0 \end{pmatrix} \right) = 6^2 = 36,$$

however this contradicts with

$$2T \left(\begin{pmatrix} 3 \\ 0 \end{pmatrix} \right) = 2(3^2 + 0) = 18$$

This definition contradicts with the one in calculus where $f(x) = ax + b$ is linear iff

$$b = 0 \quad \because \quad 0 = 0f(x) = f(0x) = f(0) = b$$

4.7 Homogeneous equations

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear function, and $\mathbf{b} \in \mathbb{R}^m$. We consider the linear equation $T\mathbf{x} = \mathbf{b}$ and solve for $\mathbf{x}$. We say $T\mathbf{x} = \mathbf{b}$ is homogeneous if $\mathbf{b} = \mathbf{0}$, and is inhomogeneous otherwise.

Suppose $\mathbf{x}_1, \dots, \mathbf{x}_r$ are solutions to the homogeneous equation $T\mathbf{x} = \mathbf{0}$.

- $\mathbf{x} = \mathbf{0}$ is a homogeneous solution.
- $\mathbf{x}_1 + \mathbf{x}_2, \lambda\mathbf{x}_1$ are homogeneous solutions for any $\lambda \in \mathbb{R}$.
- Any linear combination $\lambda_1\mathbf{x}_1 + \dots + \lambda_r\mathbf{x}_r$ is a homogeneous solution

Let $\mathbf{x} = \mathbf{x}_p$ be a solution to $T\mathbf{x} = \mathbf{b}$.

- Then $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$ is also a solution for any homogeneous solution $\mathbf{x}_h$ to $T\mathbf{x} = \mathbf{0}$.
- Every solution to $T\mathbf{x} = \mathbf{b}$ has this form.

4.8 Application to polynomials

Fix real numbers a, b, c . Are there any polynomials of degree ≤ 2 $p(x) = a_0 + a_1x + a_2x^2$ which satisfy the following conditions

$$p(1) = a, \quad p'(1) = b, \quad p''(1) = c$$

The conditions give linear conditions on the coefficients. Now

$$a = p(1) = a_0 + a_1 + a_2$$

$$b = p'(1) = a_1 + 2a_2$$

$$c = p''(1) = 2a_2$$

which can be written as

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & 1 & 2 & b \\ 0 & 0 & 2 & c \end{array} \right)$$

Chapter 5

Matrices

Recall an $m \times n$ -matrix as an array of scalars (real or complex).

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

- We abbreviate the above to $A = (a_{ij})$ and call a_{ij} the ij -th entry of A
- We also write $[A]_{ij}$ for a_{ij}
- The size of A is $m \times n$ since it has m rows and n columns
- $M_{mn}(\mathbb{R})$ denotes the set of all $m \times n$ -matrices with real entries

5.1 Matrix-vector product

Let $A = (a_{ij}) = (\mathbf{a}_1 | \mathbf{a}_2 | \dots | \mathbf{a}_n) \in M_{mn}$. Then

$$A \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \dots + x_n\mathbf{a}_n$$

5.2 Arithmetic of matrices

Matrix addition and scalar multiplication is defined to be entry-wise addition and scalar multiplication.

For $A, B \in M_{mn}(\mathbb{R})$, $\lambda \in \mathbb{R}$, the entries of $A + B, \lambda A \in M_{mn}(\mathbb{R})$ are

- $[A + B]_{ij} = [A]_{ij} + [B]_{ij}$

$$\lambda A_{ij} = \lambda[A]_{ij}$$

Note that the sum of matrices of different sizes is not defined.

- The zero matrix $\mathbf{0}$ has all entries 0, ie. $A + \mathbf{0} = A$
- The negative of $A \in M_{mn}$ is $-A := (-1)A$. Hence $A + (-A) = \mathbf{0}$
- The difference $A - B := A + (-B)$ if A, B have the same size

Example

Suppose $n = 2$ so $A = (\mathbf{a}_1 | \mathbf{a}_2), B = (\mathbf{b}_1 | \mathbf{b}_2)$

Let $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$.

$$\begin{aligned} (A + B)\mathbf{x} &= (\mathbf{a}_1 + \mathbf{b}_1 | \mathbf{a}_2 + \mathbf{b}_2) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1(\mathbf{a}_1 + \mathbf{b}_1) + x_2(\mathbf{a}_2 + \mathbf{b}_2) \\ &= x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_1\mathbf{b}_1 + x_2\mathbf{b}_2 \\ &= A\mathbf{x} + B\mathbf{x} \end{aligned}$$

5.3 Matrix multiplication

Let $A \in M_{mn}, B = (\mathbf{b}_1 | \cdots | \mathbf{b}_p) \in M_{np}$. We define the matrix product AB to be the $m \times p$ -matrix

$$AB = (A\mathbf{b}_1 | \cdots | A\mathbf{b}_p) \in M_{mp}$$

Example

$$\begin{pmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix} = \left(\begin{pmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \mid \begin{pmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} \right) = \begin{pmatrix} 1 & 2 \\ 5 & 4 \\ 9 & 6 \end{pmatrix}$$

- Matrix multiplication is associative and distributive, however not commutative

$$(A + B)^2 = (A + B)(A + B) = A^2 + BA + AB + B^2$$

5.4 Miscellaneous matrix definitions

A matrix is said to be square if it has the same number of rows as columns. The diagonal of a square matrix $A = (a_{ij})$ consists of the entries a_{ij} . The $n \times n$ matrix I is the matrix with 1's on the diagonal and 0's

elsewhere. Eg.

$$I = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} \in M_{nn}$$

5.5 Transpose

The transpose of an $m \times n$ -matrix A is the $n \times m$ -matrix A^T obtained by turning all the rows of A into columns, ie. flipping the matrix about the row i = column j diagonal.

Example

$$\begin{pmatrix} 2 & 5 & 1 \\ 1 & 3 & 2 \end{pmatrix}^T = \begin{pmatrix} 2 & 1 \\ 5 & 3 \\ 1 & 2 \end{pmatrix}$$

Let A, B be matrices and λ a scalar. The following hold when one side is defined.

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(\lambda A)^T = \lambda A^T$
- $(AB)^T = B^T A^T$

Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. Then though $\mathbf{a}\mathbf{b}$ is not defined, we can define

$$\mathbf{a}^T \mathbf{b} = (a_1 \dots a_n) \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = a_1 b_1 + \dots + a_n b_n = \mathbf{a} \cdot \mathbf{b}$$

A square matrix is symmetric if $A^T = A$ and anti-symmetric if $A^T = -A$